Registration No :

Total Number of Pages : 02 **B.Tech**
PEE4I101

4^{th} Semester Regular / Back Examination 2018-19
ELECTRICAL MACHINES-II
BRANCH : ELECTRICAL
Max Marks : 100
Time : 3 Hours
Q.CODE : F836

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

a) Why the armature winding in a DC machine is always double layer winding?
b) Distinguish between demagnetization and cross magnetization effect of armature reaction.
c) What happens if DC machine is operated at a speed below the rated speed?
d) What is the coil span to eliminate 7^{th} Harmonic in term of pole pitch?
e) Why Load angle is positive in case of alternator and negative in case of motor?
f) The resultant flux density in the air gap of synchronous generator is lowest during:
a. Open circuit b. Short circuit c. Full load d. Half Load
g) Which alternator uses damper winding, state the reason?
h) What is Short circuit ratio of Alternator and what is effect on size of alternator?
i) What are the advantages of cylindrical rotor for a turbo alternator?
j) Why the flux wave is not sinusoidal in Salient pole machine?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

a) Explain about various losses in DC machine.
b) Explain the internal and external characteristic for self and separately excited DC generator.
c) Why starter is necessary for DC motor. Explain any starting method.
d) A 400V DC motor running at 1200 r.p.m takes an armature current of 32.8Amp.The armature resistance is 0.5Ω.If the load torque increases by 25% and the flux increases by 10%, by neglecting iron and friction losses Find (i) Armature current (ii) Speed (iii) output of the machine
e) Explain universal motor. Draw speed-Load characteristics for both AC and DC and state its applications.
f) Explain the construction of alternator and write the advantages of stationary armature.
g) A 10 kVA, 440V, 1200 rpm 3 phase, Y connected alternator has armature winding resistance is (0.3+j5) Ω/phase. When generator operates at its full load and 0.8 pf lagging. Determine :
i) voltage regulation
ii) Generated emf.
h) Explain about parallel operation of alternator and state the advantages of parallel operation.
i) A 12 pole 3 phase star connected alternator has 72 slots. The flux per pole 0.88 Wb. Calculate :
i) The speed if frequency of generated EMF is 50Hz.
ii) The terminal emf for full pitch coils and 8 conductors per slot.
iii) The terminal emf if coil span is reduced to 2/3 of pole pitch.

j) What is voltage regulation of alternator? Explain synchronous impedance voltage regulation method.

k) Explain speed control methods of DC shunt motor.

l) How to get continuous unidirectional torque in synchronous motor? Explain the procedure to make it self-starting.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Discus the armature winding of D.C machine. With neat sketch show Y_F,Y_B,Y_R,Y_C,Y_S,Y_A,N_C C_S and write the relations between them for both lap and wave winding. **(16)**

Q4 Describe commutation of D.C. generator. Explain various methods for improving commutation. **(16)**

Q5 A 3300V, 3phase star connected alternator has a full load current of 100A. On short circuit afield current of 5A was necessary to produce full load current. The emf on open circuit for the same excitation was 900V. The armature resistance was 0.8Ω/phase. Determine the fullload voltage regulation for (i)0.8pf lagging (ii)0.8pf leading **(16)**

Q6 Describe the Operating principle and one starting method of three phase synchronous motor. Draw the Phasor diagram for normal, under and over excited condition. **(16)**

Registration No :

Total Number of Pages : 02 **B.Tech**

PET4I102

2nd Year Special Examination 2018-19

ELECTRICAL MACHINES & POWER DEVICES

BRANCH : ECE, ETC

Max Marks : 100

Time : 3 Hours

Q.CODE : G782

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) **(2 x 10)**

- **a)** What is the main function of dummy coil in DC machines?
- **b)** What is the function of inter pole winding and compensating winding in DC generator?
- **c)** Write the practical application of differential compound motor.
- **d)** Why in a transformer no load current is very small in comparison to induction motor?
- **e)** What is the difference between Auto transformer and two winding transformer
- **f)** What is the function of skew slot in a 3 phase induction motor?
- **g)** Draw the V-curve and inverted V curve of Synchronous motor.
- **h)** What is the advantages of Star-Delta starter over DOL starter in an Induction motor
- **i)** For improving power factor which type of Motor is used. ? What is the main function of Damper bar of an alternator??
- **j)** How induction motor can be act as induction generator?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 x 8)**

- **a)** What do you mean by Critical speed and critical resistance of DC shunt generator?
- **b)** A 4 pole DC generator runs at 750 rpm and generator runs of 240 V. The armature is wave winding and 792 conductors. If the total flux from each pole is 0.0145 Wb. What is the leakage coefficient?
- **c)** What do you mean by plugging and dynamic breaking of DC shunt generatorr?
- **d)** A 250 V DC shunt motor has shunt field resistance of 250 ohm and an armature resistance of 0.25 ohm. For a given load torque and no additional resistance included in the shunt field circuit, the motor runs at 1500 rpm drawing armature current of 20A. if a resistance is 250 ohm is inserted in series with the field, the load torque reaming same. Find out the new speed and armature current. Assume magnetization curve is linear.
- **e)** What is the condition for obtaining maximum efficiency of transformer? Derive it
- **f)** Consider a 20 KVA, 2200/220 V, 50Hz transformer. The OC/SC test results are as follows
  O.C test: 220 V, 4.2 A, 148 W (l.v side)
  S,C test: 86 V, 10.5 A, 360 W (h.v side)
  Determine the voltage regulation at 0.8 pf lagging at full load. What is the pf on Short circuit?
- **g)** Derive the expression of maximum torque under running condition in a 3 Phase induction Motor
- **h)** The power input to the rotor of a 400V 50 Hz 6 pole 3 phase induction motor is 75 KW. The rotor electromotive force is observed to make 100 complete alternations per minute. Calculate :
  (i) slip (ii) rotor speed
  (iii) rotor copper loss per phase (iv) mechanical power developed
- **i)** Derive pitch factor and distribution factor. What is the advantage of pitch factor of an alternator?

j) A 3 phase 16 pole alternator has star connected winding with 144 slots and 10 conductors per slot. The flux per pole is 0.03 Wb. Sinusoidally distributed and the speed is 375 rpm. Find the frequency rpm and phase and the line emf. Assume full pitched coil

k) Write the starting methods off Synchronous motor?

l) Discuss the torque slip characteristic of induction motor. Why this characteristics behaves as rectangular hyperbola

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 What is double field revolving theory of single phase induction motor? **(16)**

Q4 What are the different Speed control of techniques of DC shunt motor and DC series Motor? **(16)**

Q5 What is armature reaction? What are the different methods for improving commutation **(16)**

Q6 Explain the working of a 3 point starter with neat diagram **(16)**

Registration No : | | | | | | | | | | |

Total Number of Pages : 02 **B.Tech**

PEE3I103

2nd Year Special Examination 2018-19
ELECTRICAL MACHINES - I
BRANCH : ELECTRICAL
Max Marks : 100
Time : 3 Hours
Q.CODE : G317

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 **Only Short Answer Type Questions (Answer All-10)** **(2 x 10)**

a) What is the principle of operation of a transformer ?
b) What will happen if the primary of a transformer is connected to dc supply ?
c) What are the advantages of back to back test in determining the efficiency of a transformer ?
d) What do you mean by synchronous speed of three phase induction motor ?
e) Why does the rotor of a 3-phase induction motor rotate in the same direction as the rotating field ?
f) What is Cogging and Crawling of a three-phase induction motor ?
g) What is per unit system of three phase transformer ?
h) Why transformer rating in KVA ?
i) Why do we require single-phase induction machine ?
j) Why is power factor of a single-phase induction motor low ?

Part- II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 x 8)**

a) A single-phase transformer on no load takes 4.5 A at a pf of 0.25 lagging when connected to a 230 V, 50 Hz supply. The number of turns of the primary winding is 250. Calculate :
(i) the magnetizing current
(ii) the core loss and
(iii) the maximum value of flux in the core

b) Draw the circuit diagram and phasor diagram of the ideal transformer and practical transformer having no load and load condition. Write the meaning of the parameters.

c) Explain with neat sketch diagram the determination of parameters using back-to-back test.

d) An auto transformer supplies a load of 5 kW at 125 V at unity power factor. If the primary voltage is 250 V, determine :
(i) transformation ration
(ii) secondary current
(iii) primary current
(iv) number of secondary turns if total number of turns is 250
(v) power transformed inductively
(vi) power transformed conductively

e) Draw and explain with neat sketch diagram the connection of two single-phase transformer connected in Open Delta.

f) Draw and explain with neat sketch diagram the connection of two single-phase transformer Scott Connection.

g) A 3-phase, 400/200 V, Y-Y connected wound-rotor induction motor has 0.06 Ω rotor resistance and 0.3 Ω standstill reactance per phase. Find the additional resistance required in the rotor circuit to make the starting torque equal to the maximum torque.

h) A 4-pole, 3-phase, 50 Hz induction motor has a star-connected rotor. The rotor has a resistance of 0.1 Ω / phase and standstill reactance of 2 Ω / phase. The induced emf between the slip ring is 100 V. If the full load speed is 1460 rpm, calculate :

(i) the slip
(ii) the emf induced in the rotor in each phase
(iii) the rotor reactance per phase
(iv) the rotor current and
(v) rotor power factor. Assume slip rings are short circuited.

i) Draw and explain various methods of starting of three phase induction motor.

j) Write down the principle of single phase induction motor using double field revolving theory and draw the developed equivalent circuit.

k) An 8-pole, 50 Hz, 3-phase, slip-ring motor has an effective rotor resistance of 0.07 Ω / phase. It's rotor is rotating at a speed of 630 rpm. How much resistance must be inserted per phase to obtain the maximum torque at starting? Ignore the magnetizing current and the stator leakage impedance.

l) At starting, the winding of a 230 V, 50 Hz, split-phase induction motor have the following parameters:

Main winding: R= 4 Ω ; X_L = 7.5 Ω and Staring Winding : R= 7.5 Ω ; X_L = 4 Ω

Find :

(i) current I_m in the main winding
(ii) current I_s in the starting winding
(iii) phase angle between I_s and I_m
(iv) line current and
(v) power factor of the motor.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Explain the principle of operation of three-phase induction motor. Derive the equation of torque and draw the torque-slip characteristics. Explain using neat sketch circuit diagram and phasor diagram **(16)**

Q4 Explain the polarity test, open circuit test and short circuit test of the single phase transfoemer. **(16)**

Q5 a) The voltage on the secondary of a single phase transformer is 200 V when supplying a load of 8 kW at a pf of 0.8 lagging. The secondary resistance is 0.04 Ω and secondary leakage reactance is 0.8 Ω. Calculate the induced emf in the secondary. **(8)**

b) Explain the parallel operation of single–phase transformers with neat sketch diagram. What are the conditions for satisfactory parallel operation? **(8)**

Q6 a) Explain the no-load and blocked rotor test of a three phase induction motor. **(8)**

b) A 4-pole, 3-phase, 50 Hz, slip-ring induction motor rotates at 1400 rpm with slip-ring terminals short-circuited. The rotor resistance is 0.1 Ω/phase and standstill reactance / phase is 0.6 Ω. If an external resistance of 0.1 Ω/phase is added to the rotor circuit, what will be the new full-load speed? **(8)**

Registration No:

Total Number of Pages: 02

Course: B.Tech
Sub Code: REL5C003

5th Semester Regular/Back Examination: 2023-24
SUBJECT: Electrical Machines-II
BRANCH(S): EEE,ELECTRICAL
Time: 3 Hour
Max Marks: 100
Q. Code: N240
Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 **Answer the following questions:** **(2 x 10)**

- **a)** What is rotating magnetic field?
- **b)** Why short-pitch winding is preferred over full pitch winding?
- **c)** Define winding factor.
- **d)** A slip-ring induction motor runs at 290 r.p.m. at full load, when connected to 50Hz supply. The synchronous speed is 375 r.p.m. Determine the number of poles and slip.
- **e)** Define crawling and cogging of three phase induction motor.
- **f)** Why is single phase induction motor not self starting?
- **g)** Discuss few differences between single phase and three phase induction motors.
- **h)** What do you mean by armature reaction of a three phase alternator?
- **i)** An 8-pole synchronous generator is running at 750 rpm. What is the frequency? At what speed must the generator be run so that frequency shall be 25 Hz?
- **j)** Draw the V-curve and inverted V-curve at different loading conditions.

Part-II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 × 8)**

- **a)** Define the pitch factor and distribution factor and their significance in synchronous machine.
- **b)** Explain in detail the distributed and concentrated windings and how the performance of the machine can get affected by the windings construction.
- **c)** Explain the principle of operation of a 3-phase induction motor with relevant diagrams.
- **d)** A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50Hz. Calculate:
  - (i) The speed at which the magnetic field of the stator is rotating.
  - (ii) The speed of the rotor when the slip is 0.04.
  - (iii) The frequency of the rotor currents at standstill.
  - (iv) The frequency of the rotor currents when the slip is 0.03.

e) What do you mean by forward slip and backward slip in single-phase induction motor? Draw the torque- speed characteristics of single phase induction motor.

f) What are the different methods of speed control of three phase induction motor?

g) A 4-pole, 50Hz star-connected alternator has a flux per pole of 0.12Wb. It has 4 slots per pole per phase, conductors per slot being 4. If the winding coil span is 150 degree, find the EMF.

h) What do you mean by OCC and SCC of an alternator? Find the parameters from OCC and SCC with the required circuit diagram.

i) A 3-phase star connected alternator is rated at 100 kVA. On a short-circuit a field current of 50 amp gives the full load current. The EMF generated on open circuit with the same field current is 1575 V/phase. Calculate the voltage regulation at 0.8 power factor lagging. Assume armature resistance is 1.5 ohm.

j) Draw and explain the Power vs Load angle characteristics of cylindrical alternator?

k) Discuss the two reaction theory applicable to salient pole synchronous machine.

l) Justify, synchronous motor is not self starting. Explain the method of starting of synchronous motor.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Explain the torque slip characteristic of 3-phase induction motor. Derive the condition for maximum torque for an induction motor. **(16)**

Q4 A 3-phase, 4-pole, 50 Hz, induction motor has a star connected wound rotor. The rotor emf is 50V between the slip rings at standstill. The rotor resistance and standstill reactance are 0.4 ohm and 2.0 ohm respectively. Calculate, **(16)**
(i) rotor current per phase at starting when slip rings are short circuited
(ii) Rotor current per phase at starting if 50 ohm per phase resistance is connected between slip rings.
(iii) Rotor EMF when the motor us running at full load at 1440 rpm
(iv) Rotor current at full load and
(v) Rotor power factor at full load

Q5 Why are the single phase induction motor not self starting? How these can be made self starting? What are different types of Starting methods for single phase induction motor? Explain. **(16)**

Q6 Discuss and state the conditions necessary for paralleling alternators. Describe in detail the full process of synchronization of three phase alternator using dark lamp method. **(16)**

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Total Number of Pages : 02

B.Tech

PEE3I103 / PEL3I103

3rd Semester Back Examination: 2021-22

ELECTRICAL MACHINES - I

BRANCH(S): ELECTRICAL / EEE

Time: 3 Hour

Max Marks: 100

Q.Code: OF785

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 **Answer the following questions:** **(2×10)**

- **a)** What is leakage flux and how it affects the performance of the transformer?
- **b)** How does change in frequency affect the operation of a given transformer? Justify your answer.
- **c)** Why short-pitch winding is preferred over full pitch winding?
- **d)** Mention the difference between slip ring rotor and cage rotor of an induction motor?
- **e)** Why starting power factor of 3-phase induction motor is very low?
- **f)** Explain, why single-phase induction motor is not self-starting type.
- **g)** Why shell type 3-phase transformer is used in large power transforming applications?
- **h)** What is the role of tertiary winding in 3 phase transformers?
- **i)** Why the core loss is neglected in case of short circuit test of 1-pahse transformer.
- **j)** Mention the most common connections types are used in case of 3 phase transformer.

Part-II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6×8)**

- **a)** A 3 phase, 6,600/415 V, 2000kVA, transformer has a per unit resistance of 0.02 and a per unit leakage reactance of 0.1. Calculate the Cu-loss and regulation at full-load, 0.8 p.f lag.
- **b)** Mention the advantages and disadvantages of a bank of 3, 1-phase transformers to single 3-phase transformers.
- **c)** In case of an 8-pole induction motor the supply, frequency is 50 Hz and the shaft speed is 735 r.p.m. Calculate i) Synchronous speed ii) Slip speed iii) Percentage slip
- **d)** Explain the procedure for no-load and blocked rotor tests on a 3-phase induction motor.
- **e)** Describe the procedure for conducting Sumpner's test on a pair of transformers with neat circuit diagram. Why polarity test important before doing any connection in secondary side.
- **f)** Explain torque-speed characteristics of an induction motor.

g) What do you mean by vector group of transformers? Explain about Yd11, Dy11, and Yz11.

h) Explain in brief about the double field revolving theory for operation of single-phase induction motor.

i) If P_g is the air gap power, then prove that rotor ohmic loss is sP_g.

j) Explain with neat sketch for 1-phase transformer polarity test.

k) Mention the advantages and disadvantages of auto transformer over two winding transformers.

l) Explain with neat sketch the star-delta starter and derive the expression for starting current and torque

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 A 3-phase Δ-Δ bank consists of three 25 kVA, 3,300 / 300 V transformers and supplies a load of 50 kVA. After removing one transformer, determine the following for Y-Y connection. **(16)**

i) kVA load carried by each transformer,

ii) Percentage of rated load carried by each transformer,

iii) Total kVA rating.

Q4 With neat phasor diagram explain the voltage regulation of three-phase transformer and give the applications of various types of three phase transformers **(16)**

Q5 Explain various speed control methods of 3-phase induction motor with neat diagrams **(16)**

Q6 Draw the phasor diagram of 3-phase induction motor at standstill and at full load slip s and explain each of the phasor diagrams in details. **(16)**

Registration No : | | | | | | | | | | |

Total Number of Pages : 03 **B. Tech**

REE4C002 / REL4C002

4^{th} Semester Regular / Back Examination: 2021-22

ELECTRICAL MACHINES-I

BRANCH(S): EEE / ELECTRICAL

Time : 3 Hour

Max Marks : 100

Q. Code : J451

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : **(2 × 10)**

- **a)** What is fringing effect and what causes it?
- **b)** Why the pole shoes are laminated whereas the pole core not?
- **c)** Why the air-gap between the yoke and armature in a dc motor is kept small?
- **d)** How a DC shunt generator is self-protecting from short circuit.
- **e)** How a DC motor adjust itself to draw the required current to full fil the load demand?
- **f)** Why the conductor used in NVR is thinner and in OLR it is thicker?
- **g)** Why DC series motor is never operated at no-load and light load conditions?
- **h)** Two transformers of same type, using the same grade of iron and conductor materials, are designed to work at the same flux and current densities, but the linear dimensions of the core of one is three times of the other and the conductor diameter is double. The ratio of KVA rating of the two transformers will be______?
- **i)** A single-phase transformer has percentage resistance of 2% and constant loss of 1%. What will be the maximum possible efficiency of this transformer?
- **j)** Why the no-load current of a transformer is non-linear?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 × 8)**

- **a)** Explain, how the B-H characteristics of a magnetic material can be plotted and show the effect of magnetic material on its shape?
- **b)** Derive the general expression for torque developed by a rotating element?
- **c)** Explain the effect of armature reaction of a DC machine?
- **d)** Describe the working of an interpole?
- **e)** Describe the process of voltage build-up in a shunt generator?
- **f)** Explain back-to-back test of two single phase transformer?
- **g)** Explain how a transformer connected in Yd_{11} can be successfully operated parallel with another connected in Dy_1?
- **h)** Derive the expression for copper saving in an autotransformer compared to a two winding transformer?
- **i)** A three-winding transformer is connected to an AC voltage source as shown in the figure. The numbers of turns are as follows: N_1=100, N_2=50, N_3=50. If the magnetizing current is neglected, and the currents in two windings are I_2= 2 ∠30° A and I_3= 2∠150° A, then what is the value of the current I_1 in Ampere?

j) A 200/400 V, 50 Hz, two-winding transformer is rated at 20 kVA. Its windings are connected as an auto-transformer of rating 200/600 V. A resistive load of 12 Ω is connected to the high voltage (600 V) side of the auto-transformer. What is the value of equivalent load resistance (in Ohm) as seen from low voltage side?

k) A three-phase step down transformer is connected to 3300V supply. The turns ratio per phase is 12 and it draws a current of 5 A from the mains. Calculate the secondary line voltage and line current if transformer is

- **(a)** Delta-star connected.
- **(b)** Star-delta connected.

l) Show the connections and phasor diagram of

- **(a)** Dy_{11}
- **(b)** Dz_0

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 a) A 8 pole DC shunt generator has 778 wave connected armature conductor running 500 RPM, supplies a load of 12.5 Ω at a terminal voltage of 250V determined armature current, generated emf and flux per pole if armature and field circuit resistance are 0.24 Ω and 250 Ω respectively. **(8)**

b) A DC motor has the following specifications: 10 hp, 37.5 A, 230 V; flux/pole = 0.01 Wb, number of poles = 4, number of conductors = 666, number of parallel paths = 2. Armature resistance is 0.267 Ω. The armature reaction is negligible and rotational losses are 600 W. The motor operates from a 230 V DC supply. If the motor runs at 1000 rpm, the output torque produced (in Nm) is? **(8)**

Q4 a) A belt driven 60KW shunt wound generator running at 500rpm is supplying full load to a bus bar at 200V. At what speed will it run if the belt breaks and the machine continued to run taking 5KW from the bus bar. Assume armature and shunt field resistance are 0.1Ω and 100 Ω respectively. Consider 1V drop per brush. **(8)**

b) A 250V, D.C. shunt motor has an armature resistance of 0.5 Ω and field resistance of 250 Ω. When driving a constant torque load at 600 rpm, the motor draws 21A. What will be the new speed of the motor if an additional 250 Ω resistance is inserted in series in the field circuit? **(8)**

Q5 a) The constants of a single phase 50 Hz, 20kVA, 2200/220 V transformer are as follows **(8)**

h.v. side: r_1=0.21Ω, x_1=3.4 Ω,

l.v. side : r_2=0.006 Ω, x_2=0.022 Ω

a) Find the equivalent circuit parameters referred to both hv and v side.

b) Calculate full load ohmic loss.

c) Find secondary terminal voltage and efficiency with a load impedance of 5+j4 Ω connected on lv side with rated excitation.

b) Given that the full load copper losses are exactly twice the iron losses in a 50 kVA transformer, and that the quarter load efficiency is 96.5%, calculate the full load efficiency at unity power factor. Also, determine the maximum efficiency of the transformer? **(8)**

Q6 **a)** A single-phase, 22 kVA, 2200 V/ 220 V, 50 Hz, distribution transformer is to be connected as an auto-transformer to get an output voltage of 2420 V from 220V input. Determine its kVA rating. If the full load efficiency of the transformer is 97% at UPF, then determine the efficiency at the same load, when connected as an autotransformer? **(8)**

b) A Scott-connected transformer is fed from a 6,600-V, 3-phase network and supplies two single-phase furnaces at 100 V. Calculate the line currents on the 3-phase side when the furnaces take 400 kW and 700 kW respectively at 0.8 power factor lagging. **(8)**

Registration No :

Total Number of Pages : 02

B.Tech
PEE3I103

3rd Semester Back Examination 2019-20
ELECTRICAL MACHINES - I
BRANCH : ELECTRICAL
Max Marks : 100
Time : 3 Hours
Q.CODE : HB891

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.
The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) **(2 x 10)**

- **a)** Under what condition transformer will operate at leading power factor.
- **b)** Give the expression of cupper savings in auto transformer as compare to two winding transformers.
- **c)** Why the rotor slots of a 3-phase induction motor are skewed?
- **d)** Explain why the no load current of an induction motor is much higher than that of an equivalent transformer.
- **e)** What is the function of capacitor in single phase induction motor?
- **f)** Why it is mandatory to use low power factor watt meter during no-load test of 3 phase induction motor?
- **g)** What is leakage reactance drop? Explain.
- **h)** Mention the different types of three phase transformer connection.
- **i)** Why all-day efficiency is lower than commercial efficiency?
- **j)** What are the conditions for parallel operation of a transformer?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 x 8)**

- **a)** Draw the equivalent circuit diagram of single-phase transformer referred to Low voltage side and explain in details of each components.
- **b)** Where the damper windings are located? What are their functions?
- **c)** A 120kVA, 6600/400V, Y/Y, 3-phase, 50Hz transformer has a iron loss of 1600W. The maximum efficiency occurs at ¾ full loads. Find the efficiency of the transformer at
  (i) Full load and 0.8 pf
  (iii) The maximum efficiency at unity pf.
- **d)** Mention the different phasor group connections for 3 phase transformer and Explain any one of the group.
- **e)** Derive an expression for the currents and load shared by two transformers connected in parallel supplying a common load when no load of these are equal.
- **f)** Obtain the equivalent circuit of a 200/400V,50Hz, single phase transformer from the following test data :
  OC test: 200V,0.7A,70W on LV side
  SC test: 15V,10A,85W on HV side

- **g)** A 3-phase induction motor has a starting torque of 150% and maximum torque of 250% of full load torque. Neglect stator resistance and assume constant rotor resistance. Compute (a) Slip at maximum torque (b) Full load slip
- **h)** Explain effect of unbalance voltages & frequency variation on operation of induction motor.
- **i)** Explain torque-slip characteristic of three-phase induction motor.
- **j)** Explain how a rotating magnetic field is produced in a three-phase induction motor.
- **k)** Draw and explain the Full load Phasor Diagrams of Single-Phase transformer for lagging, leading and Unity power factor loads
- **l)** Show that open delta connection has a KVA rating of 58% of the rating of the normal delta-delta connection. Also list the limitations of open delta connection

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Draw and explain the torque/slip curves of a three-phase induction motor for different values of rotor resistance. **(16)**

Q4 Explain the different speed control methods of phase wound induction motor. **(16)**

Q5 A100 kVA, 6.6kV/415V, single phase transformer has an effective impedance of (3+8j) Ω referred to HV side. Estimate the full load voltage regulation at 0.8 pf lagging and 0.8 leading pf. **(16)**

Q6 Derive the expression for air gap power, mechanical power developed and internal power for 3 phase induction motor. **(16)**

Registration no: | | | | | | | | | | |

Total Number of Pages: **B.TECH**

PCEE4203

4^{TH} Semester Back Examination 2016-17

ELECTRICAL MACHINE – I

BRANCH(S): EEE, ELECTRICAL

Time: 3 Hours

Max Marks: 70

QCODE: Z326

Answer Question No.1 which is compulsory and any five from the rest.

The figures in the right hand margin indicate marks.

Q1 **Answer the following questions:** **(2 x 10)**

a) Between DC shunt and separately excited generators, whose terminal voltage is high? Justify.

b) How the direction of rotation of a DC series motor can be reversed?

c) What is the function of interpoles and how are interpole winding connected?

d) Why dc series motor should never be started without any mechanical load?

e) What are the uses of DC compound generators?

f) Draw the torque speed characteristic of three phase induction motor.

g) What is the advantage of slip ring induction motor over squirrel cage induction motor?

h) Why starting current of Induction motor is higher than the transformer?

i) Draw the phasor diagram of a lossless transformer at no load.

j) A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50Hz, calculate the speed of rotor and rotor flux with respect to stator flux.

Q2

a) Discuss various losses and power stages of DC Generator and hence derive the condition for maximum efficiency. **(5)**

b) Explain the voltage build up process of DC shunt generator and discuss the importance of Critical speed and critical resistance. **(5)**

Q3

(a) A 250/500V single phase transformer gave the following test results **(5)**
SC Test(HV side): 20V, 12A, 100W
OC Test(LV Side): 250V, 1A, 80W
Determine the circuit constants, insert these on the equivalent circuit diagram and calculate the efficiency when the output is 10A at 500V and 0.8 power factor lagging.

(b) With neat diagram, explain how Sumpers back to back test is used to find the efficiency of a transformer. **(5)**

Q4

a) A 4 pole, 32 conductor, lap-wound dc shunt generator with terminal voltage of 200volts delivering 12 amps to the load has armature resistance 2 ohm and field resistance of 200ohms. It is driven at 1000rpm. Calculate the flux per pole in the machine. If the machine has to be run as a motor with the same terminal voltage and drawing 5A from the mains, find the speed of the motor. **(5)**

b) Compare the torque-current and speed-current characteristics of DC shunt and Series Motors and hence give their uses. **(5)**

Q5 **a)** Derive the expression for voltage regulation of a single phase transformer. Is it possible to have zero voltage regulation? Jusify. **(5)**

b) A short shunt dc compound generator supplies 200A at 100V. The resistance of armature, series field and shunt field windings are 0.04, 0.03 and 60 ohms respectively. Find the emf generated if voltage drop per brush is 1V. Also find the emf generated if same machine is connected as a long shunt machine. **(5)**

Q6 **a)** A 6 pole, 50Hz, 3 phase induction motor running on full load with 4% slip develops a torque of 149.3 Nm at its pulley rim. The friction and windage losses are 200W and the stator Cu and iron losses equal to 1620W. Calculate (i) output power (ii) the rotor copper loss (iii) the efficiency at full load. **(5)**

b) What are different speed control methods of 3 phase induction motor? **(5)**

Q7 **a)** Draw and explain the equivalent circuit and phasor diagram of 3 phase induction motor. How it is different from transformer phasor diagram? **(5)**

b) For the 20KVA, 2400/240V two winding transformer is reconnected as an autotransformer. Find the input and output rated current and KVA rating of the auto transformer for maximum voltage gain from input to output. Show the winding connection of the auto transformer. **(5)**

Q8 **Write short notes on any two:** **(5 x 2)**

a) Starting of 3-pahse Induction motor
b) Armature reaction in DC machine
c) Crawling

Registration No: | | | | | | | | | | |

Total Number of Pages: 03 **B.TECH**

PEE4I101

4th Semester Regular Examination 2016-17
ELECTRICAL MACHINE - II
BRANCH: ELECTRICAL
Time: 3 Hours
Max Marks: 100
Q.CODE: Z434
Answer Part-A which is compulsory and any four from Part-B.
The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: **(2 x 10)**

a) In a dc machine the shape of armature mmf distribution in space is
(A) Sinusoidal(B) Triangular (C) Rectangular(D) Trapezoidal

b) In a loaded dc generator, if the brushes are given a shift from the interpolar axis in the direction of rotation, then the commutation will
(A) Improves with fall in V_t (B) Improves with rise in V_t
(C) Deteriorate with fall in V_t(D) Deteriorate with rise in V_t

c) The no load terminal voltage of a dc shunt generator is 220V. If the direction of rotation is reversed the voltage to which it build up is:
(A) 220 V (B) -220 V(C) 0 V (D) 22 V

d) The dc motor, which can provide zero speed regulation at full load without any controller is
(A) series (B) shunt
(C) cumulative compound (D) differential compound

e) A 4 point starter is used to start and control the speed of a
(A) dc shunt motor with armature resistance control
(B) dc shunt motor with field weakening control
(C) dc series motor (D) dc compound motor

f) A field excitation of 20 A in a certain alternator results in an armaturecurrent of 400 A in short circuit and a terminal voltage of 3464 V on opencircuit. The magnitude of the internal voltage drop within the machine at a
load current of 200 A is
(A) 1 V (B) 10 V(C) 100 V (D) 1000 V

g) A 3 phase, 50Hz, 4 pole alternator has a rotating armature. What will be the speed of rotating magnetic field w.r.t the stator?
(A) 1500rpm (B) -1500rpm (C) 0rpm(D) 3000 rpm

h) An over excited synchronous motor operates at_____p.f. by _____reactive power.
(A) Lagging, delivering (B) Lagging, consuming
(C) Leading, delivering (D) Leading, consuming

i) The maximum power developed in the synchronous motor will depend on
(A) rotor excitation only
(B) maximum value of coupling angle
(C) supply voltage only
(D) rotor excitation supply voltage and maximum value of coupling angle.

j) The type of machine having highest no load current is
(A) Transformer (B) DC shunt motor
(C) 3 phase Synchronous motor (D) 3 phase Induction motor

Q2 Answer the following questions: **(2 x 10)**

a) Why does the terminal voltage fall more rapidly in a self excited shunt generator than in a separately excited dc generator?

b) How should the main pole tips be constructed to minimize the effect of armature reaction?

c) Four terminals of a dc shunt machine are available, but these are unmarked. How would you identify field and armature terminals?

d) How does a dc motor adjust itself to match the mechanical load?

e) An alternator has the same voltage, frequency and phase sequence as that of the infinite bus. Is it sufficient to put ON the synchronization switch now? Give reasons for your answer.

f) For the same kVA load, efficiency of an alternator at leading p.f. is more than at lagging p.f. Explain.

g) A synchronous motor is supplying a constant load. In order to stop the motor, it is essential to switch-off the ac supply first then its field current. Explain the reason.

h) Two 20-MVA, 3-phase alternators operate in parallel to supply a load of 35 MVA at 0.8 pf. Lagging. If the output of one machine is 25 MVA at 0.9 lagging, what is the output and pf. of the other machine?

i) Why the air gap length in a synchronous machine is higher compared to an induction machine?

j) What is auniversal motor? Mention two of its application.

Part – B (Answer any four questions)

Q3 a) A 6 pole lap wound generator has 240 coils of 2 turns each. Resistance of 1 turn is 0.03 Ω the armature is 50 cm long and 40 cm diameter. Air gap flux density of 0.6T is uniform over pole shoe. Each pole subtends and angle of 40^0 mechanical. For a speed of 1200rpm determined Eg, internal torque developed and terminal voltage for a load current of 40 A. **(10)**

b) Explain the process of voltage buildup in dc shunt generator. **(5)**

Q4 a) A 230V DC shunt motor has field resistance of 115 Ω and armature circuit resistance of 0.5 Ω. At no load, the motor runs at 1000 rpm, with armature current of 4A and rated field flux. **(10)**
(i) For a developed torque of 80Nm. compute armature current and speed.
(ii) If it is desired that motor develops 8 kW at 1250 rpm, determine the value of external resistance that must be inserted in series with the field winding.

b) Explain T~Ia, N~Ia and T~N characteristics of dc shunt motor. **(5)**

Q5 **a)** A 250V, 20Kw shunt motor running at 1500 rpm has a maximum efficiency of 85%, when delivering 80% of its rated output. The shunt field resistance is 125Ω. Determine efficiency, output torque and speed when it draws 100A from mains. **(10)**

b) Calculate the rms value of the induced emf per phase of a 6-pole, 3-phase, 50 Hz star connected alternator with 4 slots/pole/phase and 8 conductors/slot in two layers. The coil is shorted by 2 slots. The flux per pole has fundamental component of 0.012 wb, a 20% third harmonic and 8% fifth harmonic. **(5)**

Q6 **a)** A 3-phase, 17.32kVA, 400 V, star connected alternator is delivering rated load at 400 V and at pf 0.8 lagging. Its synchronous impedance is 0.2+j2 Ω per phase. Find the load angle of operation and voltage regulation. Also calculate the p.f. for which the voltage regulation becomes zero. **(10)**

b) A 5kVA, 220V, 3-phase, star connected, salient pole alternator with direct and quadrature axes reactance of 12Ω and 7Ω respectively, delivers full load current at 0.8 power factor lagging. Calculate the excitation voltage and load angle neglecting armature resistance. Also calculate the reactance power developed by it. **(5)**

Q7 **a)** A 3-phase star connected alternator has synchronous impedance of 1+j 10 Ω per phase. It is operating at a constant voltage of 6.6kV and its field current is adjusted to give an excitation voltage of 6.4kV. Find the power output, armature current and pf under the condition of (i) maximum power output and (ii) maximum power input. **(10)**

b) Explain how the direct and quadrature axis reactance of salient pole synchronous machine can be determined? **(5)**

Q8 **a)** Two exactly similar turbo alternators are rated at 25MW each. They are running in parallel. The speed load characteristics of the driving turbines are such that the frequency of alternator 1 drops uniformly from 50Hz.on no-load to 48Hz on full load, and that of alternator 2 from 50Hz to 47.5Hz.How will the two machines share a load of 40MW and maximum load that can be delivered by both without overloading either of them? **(10)**

b) Find the expression for $tan\delta$for a salient pole synchronous generator working at lagging power factor. **(5)**

Q9 **a)** Explain various methods for starting a three phase synchronous motor. **(10)**

b) A universal series motor has resistance of 35ohms and an inductance of 0.5H. When connected to 230V dc supply and loaded to take 1A it runs at 2000rpm. Determine the speed, torque and power factor, when connected to a 230V, 50Hz ac supply and loaded to take the same current as of dc supply. **(5)**

Registration No: | | | | | | | | | | |

Total Number of Pages: 03

B.TECH
PEE3I103

3rd Semester Regular Examination 2016-17
ELECTRICAL MACHINE - I
BRANCH: ELECTRICAL
Time: 3 Hours
Max Marks: 100
Q.CODE:Y531

Answer Part-A which is compulsory and any four from Part-B.
The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: **(2 x 10)**

a) In a transformer, zero voltage regulation at full load is
(A) not possible
(B) possible at unity power factor load
(C) possible at leading power factor load
(D) possible at lagging power factor load

b) The slip of an induction motor normally does not depend on
(A) rotor speed (B) synchronous speed
(C) shaft torque (D) core-loss component

c) For a single phase capacitor start induction motor which of the following statements is valid?
(A) The capacitor is used for power factor improvement
(B) The direction of rotation can be changed by reversing the main winding terminals
(C) The direction of rotation cannot be changed
(D) The direction of rotation can be changed by interchanging the supply Terminals

d) A three-phase 440 V, 6 pole, 50 Hz, squirrel cage induction motor is running at a slip of 5%. The speed of stator magnetic field to rotor magnetic field and speed of rotor with respect of stator magnetic field are
(A) zero, −5 rpm (B) zero, 955 rpm
(C) 1000 rpm, −5 rpm (D) 1000 rpm, 955 rpm

e) Distributed winding and short chording employed in AC machines will resultin
(A) increase in emf and reduction in harmonics
(B) reduction in emf and increase in harmonics
(C) increase in both emf and harmonics
(D) reduction in both emf and harmonics

f) A 500 kVA, 3-phase transformer has iron losses of 300 W and full load copper losses of 600 W. The percentage load at which the transformer isexpected to have maximum efficiency is
(A) 50.0% (B) 70.7%(C) 141.4% (D) 200.0%

g) A single phase transformer has a maximum efficiency of 90% at full loadand unity power factor. Efficiency at half load at the same power factor is
(A) 86.7% (B) 88.26%(C) 88.9% (D) 87.8%

h) The type of single-phase induction motor having the highest power factorat full load is
(A) shaded pole type (B) split-phase type
(C) capacitor-start type (D) capacitor-run type

i) It is desired to measure parameters of 230 V/115 V, 2 kVA, single-phase transformer. The following wattmeters are available in alaboratory:
*W*1 : 250 V, 10 A, Low Power Factor   *W*2 : 250 V, 5 A, Low Power Factor
*W*3 : 150 V, 10 A, High Power Factor   *W*4 : 150 V, 5 A, High Power Factor

The Wattmeters used in open circuit test and short circuit test of the transformer will respectively be
(A) *W*1 and *W*2 (B) *W*2 and *W*4   (C) *W*1 and *W*4   (D) *W*2 and *W*3

j) The direction of rotation of a 3-phase induction motor is clockwise when itis supplied with 3-phase sinusoidal voltage having phase sequence A-B-C.For counter clockwise rotation of the motor, the phase sequence of the powersupply should be
(A) B-C-A (B) C-A-B(C) A-C-B (D) B-C-A or C-A-B

Q2 Answer the following questions: **(2 x 10)**

a) Why the voltage regulation of a transformer is zero or negative for leading power factor load?

b) A single phase transformer has a hysteresis and eddy current loss of 150W and 100W respectively when supplied from 250V, 50Hz. What will be the corresponding losses when supplied from 220V, 30Hz?

c) Draw the phasor diagram of a transformer supplying power at full load 0.8 leading power factor.

d) A 3-phase I.M. has slots/pole/phase=5. If the coil span=13 slots, determine the winding factor.

e) Explain the advantage of using tertiary winding in 3 phase transformer?

f) Three single phase transformer connected in Dd0, delivering full load, if one of the transformer is taken out of operation, find the % overloading of each transformer?

g) Explain why an induction motor at no load operates at a very low power factor?

h) Why you need a starter to start a poly phase induction motor?

i) Draw the torque slip characteristics of a single phase Induction motor?

j) Explain how the speed of a single phase induction motor can be controlled?

<u>Part – B (Answer any four questions)</u>

Q3 a) A 5KVA, 2200/220 V, single phase transformer has the following parameters R_1=3.4Ω, R_2=0.028 Ω, X_1=7.2 Ω and X_2=0.06 Ω. Determine a) the input current and power factor, b) Terminal voltage when a load of 4+j5 Ω is connected (c) Equivalent circuit refer to both side. **(10)**

b) Derive the expression for voltage regulation for a single phase transformer. **(5)**

Q4 a) A single-phase, 50 kVA, 250 V/500 V two winding transformer has an efficiency of 95% at full load, unity power factor. If it is re-configured as a500 V/750 V auto-transformer, calculate its efficiency at rated load and unity power factor. Also show the current distribution and calculate the KVA output for both additive and subtractive polarity connection. **(10)**

b) Explain the procedure for conducting an open and short circuit test in a laboratory of a single Phase transformer by drawing a neat circuit diagram. Also explain the information is obtained from this test to draw the equivalent circuit of the single phase transformer ? **(5)**

Q5 a) Show that in a scott connected transformer if the load is balance then the three phase input current are also balanced. **(5+5)**
Two single phase furnaces are supplied at 250V from a 6.6kV, 3 phase system through a pair of Scott connected transformers. If the load on the main transformer is 85kW at 0.9p. f (lag) and on the teaser transformer is 69kW at 0.8 p.f (lag) Find the values of line current on the three phase side? (Neglect the magnetizing and core loss currents in the transformer).

b) Explain phasor group Dy1 & yZ11 with reference to three phase transformer showing suitable clock diagram? Also show the connection and phasor diagram for the above group. **(5)**

Q6 a) Explain different methods available for controlling speed of 3 phase induction motor? **(10)**

b) A 3 phase induction motor having a 6pole star connected stator winding runs at 240V , 50HZ supply .The rotor resistance and stand still reactances are 0.12 Ω and 0.85 Ω per phase .The ratio of stator to rotor turns is 1.8 and full load slip is 4%. Calculate the developed torque at full load and maximum torque **(5)**

Q7 a) A 3 phase induction motor having a 6pole star connected stator winding runs at 240V, 50HZ supply. The rotor resistance and stand still reactance are 0.12 Ω and 0.85 Ω per phase. The ratio of stator to rotor turns is 1.8 and full load slip is 4%. Calculate the developed torque at full load and maximum torque? Also calculate the slip at which maximum efficiency occurs? **(10)**

b) Explain various method of starting a three phase Induction motor. **(5)**

Q8 a) Explain double field revolving theory? **(10)**

b) Draw the circuit model of a single phase induction motor, and explain how the parameters can be calculated from no load and blocked rotor test. **(5)**

Q9 Write short notes on any three: **(5X3)**

a) Back to Back Test.

b) Starting of single phase Induction motor.

c) Crawling

d) Open Delta (V) Connection

Registration no: | | | | | | | | | | |

Total Number of Pages: 2 **B.TECH**
PCEE4203

4th Semester Regular / Back Examination 2015-16
ELECTRICAL MACHINE - I
BRANCH(S): EEE,ELECTRICAL
Time: 3 Hours
Max Marks: 70
Q.CODE: W141

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: **(2 x 10)**

a) Give reasons, if a dc shunt generator fails to build up the voltage at its rated speed.

b) A long shunt compound generator is cumulatively compounded. With the supply terminals unchanged the machine is now runs as compound motor. If shunt field is stronger than series field. Is the motor differentially or cumulatively compounded? Justify your answer. Comment on the direction of rotation.

c) A dc shunt generator while running at 1500 rpm develops a terminal voltage of 250V at no load. If it operated as motor on no load with the same terminal voltage, then what speed it may run, 1500 rpm, more than that or less than that.

d) Explain Why the wire used in no load release coil is thinner where as it is thicker in over load release coil?

e) Because of the dc magnetic field, there is no hysteresis loss in the stator. Can it be same for the rotor also? Justify your answer.

f) DC Shunt generator is self protecting from short circuit. Justify.

g) Why the voltage regulation of a transformer is zero or negative for leading power factor load?

h) A single phase transformer has a hysteresis and eddy current loss of 100W and 75W respectively when supplied from 250V, 50Hz. What will be the corresponding losses when supplied from 220V, 30Hz?

i) Draw the phasor diagram of a transformer supplying leading power factor load?

j) A 3-phase, 4-pole, 50Hz Induction motor is running at 5% slip. Calculate the speed of the rotor magnetic field with respect to both rotor and stator.

Q2 **a)** Explain the Internal and External characteristics of a dc Shunt generator. **(5)**

b) A belt driven 50KW shunt wound generator running at 500rpm is supplying full load to a bus bar at 200V. At what speed will it run if the belt breaks and the machine continued to run taking 5KW from the bus bar. In generating mode armature reaction weakens the field by 5%. Consider armature and shunt field resistance are 0.1Ω and 100 Ω respectively and 1V drop per brush. **(5)**

Q3 a) Two shunt generators operating in parallel have linear characteristics one machine has a terminal voltage of 260V on no load and 220V at a load current of 30A, the other machine has a voltage of 270V on no load and 220V at a current of 40A. Calculate the output current, power of each machine and Bus voltage when supplying a Load resistance of 10 Ω **(5)**

b) Explain how the speed of a dc shunt motor can be controlled by controlling the flux also mention its advantages and disadvantages. **(5)**

Q4 a) Explain the need of interpoles and compensating windings in a dc motor. **(5)**

b) A 15kW, 4-pole, 230V dc shunt motor has an armature resistance of 0.2 Ω and 882 lap connected armature conductors. It delivers its rated load at 1150 rpm, while drawing an armature current of 73A, at a field current of 1.6A. Calculate the developed torque. Also find the new operating speed if the field flux reduces by 80% for the same torque developed. **(5)**

Q5 a) A 250V, 20Kw shunt motor running at 1500 rpm has a maximum efficiency of 85%, when delivering 80% of its rated output. The shunt field resistance is 125Ω. Determine efficiency and speed when it draws 100A from mains. **(5)**

b) A 5KVA, 2200/220 V, single phase transformer has the following parameters R_1=3.4Ω, R_2=0.028 Ω, X_1=7.2 Ω and X_2=0.06 Ω. Determine a) the input current and power factor, b) Terminal voltage when a load of 4+j5 Ω is connected. **(5)**

Q6 a) Obtain the approximate equivalent circuit of a given 200/2000V single phase 30kVA transformer having following test results : **(5)**

O.C Test : 200V 6.2A 360W on LV side
S.C Test : 75V 18A 600W on HV side

Also calculate the maximum efficiency at upf.

b) A 20kVA, 2200/220 V, two winding transformer is to be used as an auto transformer, with an input voltage of 2200V. Show the current distribution and calculate the KVA output for both additive and subtractive polarity connection. **(5)**

Q7 a) A 10 kW, 4 pole, 50Hz 3-phase IM has a full load slip of 0.03. Mechanical and stray load losses at full load are 3.5% of output power. Compute: **(5)**

Air gap power
Full load internal torque.
Full load rotor ohmic loss.

b) Explain various method of starting a 3-phase Induction Motor. **(5)**

Q8. Answer any two **(5 x 2)**

a) Give a comparison study of speed torque characteristics of shunt series and compound motor.

b) Hopkinson's Test

c) Induction Generators

d) Process of voltage build up in a DC Shunt generator.

Registration No :

Total Number of Pages : 02

B.Tech
PEE3I103

3rd Semester Regular / Back Examination 2018-19
ELECTRICAL MACHINES – I
BRANCH : ELECTRICAL
Time : 3 Hours
Max Marks : 100
Q.CODE : E933

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.
The figures in the right hand margin indicate marks.

Part- I

Q1 Short Answer Type Questions (Answer All-10) **(2 x 10)**

a) Classify the AC machines?
b) Draw the phasor diagram for a practical transformer with lagging power factor load.
c) What do understand by an ideal transformer?
d) Explain why the primary mmf must be equal and opposite to the secondary mmf in an ideal transformer.
e) Why per unit system of measurement is required in machine performance analysis?
f) What is main drawback of an autotransformer as compared to an ordinary transformer? What is the application of auto transformer?
g) How third harmonic component in a power transformer can be eliminated without filter circuit?
h) A 208-V, 60-Hz, 4-pole, three-phase induction motor has a full-load speed of 1755 rpm. Calculate (a) its synchronous speed, (b) the slip, and (c) the rotor frequency.
i) Explain why aninduction motor cannot operate at its synchronous speed.
j) What would happen If the rotor of the induction motor is driven faster than synchronous speed? Draw the torque-speed characteristic showing the statement.

Part- II

Q2 Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 x 8)**

a) The magnetization current in apractical transformer is not sinusoidal. Explain the statement with waveform.
b) Draw and explain the exact and approximate equivalent circuit of a transformer.
c) The available power out of the open-delta bank is only 57.7 percent of the original bank's rating. Justify.
d) Draw the experimental set-up for open and short circuit test of single phase transformer.
e) A 2000/200 V, 20kVA transformer is connected as a step-up auto-transformer (2000/2200V). Calculate its kVA rating, kVA transferred inductively, conductively and its efficiency at full load 0.8 p.f.
f) Describe the no-load test, blocked-rotor test of an induction motor.
g) Draw the flow diagram for power input to output including losses at various stages of the three phase induction motor.
h) Discuss about the crawling and cogging of induction motor.
i) Draw the experimental set-up of a back-to back connection of two single phase transformers. Why this is done so?
j) Derive the expression for copper saving of an auto-transformer as compared to ordinary transformer of same rating.
k) Develop the criteria for the maximum torque developed by induction motor during running.
l) Discuss the double field revolving theory. Draw the equivalent circuit for a single-phase induction motor considering both forward and backward rotor branches at rest.

Part-III

Long Answer Type Questions (Answer Any Two out of Four)

Q3 Define Voltage regulation and all-day efficiency of transformer. **(16)**

A 15-kVA, 2300/230-V transformer is to be tested to determine its excitation branch components, its series impedances, and its voltage regulation. The followingtest data have been taken from the primary side of the transformer.

Open-circuit test	Short-circuit test
$V_{OC} = 2300\,V$	$V_{SC} = 47\,V$
$I_{OC} = 0.21\,A$	$I_{OC} = 6.0\,A$
$P_{OC} = 50\,W$	$P_{SC} = 160\,W$

a) Find the equivalent circuit of this transformer referred to the high-voltage side.
b) Find the equivalent circuit of this transformer referred to the low-voltage side.
c) Calculate the full-load voltage regulation at 0.8lagging power factor, 1.0 power factor, and at 0.8 leading power factor.
d) Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8lagging, 1.0, and 0.8leading.
e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Q4 Draw the exact equivalent circuit to show the stator, rotor and load resistance of a three phase induction motor. **(16)**

A 10-hp, 4-pole, 440-V, 60-Hz, Y-connected, three-phase induction motor runs at 1725 rpm on full load. The stator copper loss is 212 W, and the rotational loss is 340 W. Determine (a) the power developed, (b)the air gap power, (c) the rotor copper loss, (d) the total power input, and (e) the efficiency of the motor. What is the shaft torque?

Q5 What are the different methods of speed control for three phase induction motor. Discuss the speed control by variable frequency control methods for considering variable mechanical load. Draw the torque speed characteristics for frequency control method. **(16)**

Q6 Write down the construction and principle of operation of three phase transformer. Draw the different vector group connections and their phasor diagram of three phase transformer. **(16)**

Registration No :

Total Number of Pages : 02

B.Tech
PCEL4302

5th Semester Back Examination 2018-19
ELECTRICAL MACHINES - II
BRANCH : EEE, ELECTRICAL
Time : 3 Hours
Max Marks : 70
Q.CODE : E211

Answer Question No.1 which is compulsory and any FIVE from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions : **(2 x 10)**

a) List the necessary conditions for parallel operation of three phase alternators.
b) Enlist the parameters required to get V- curves.
c) Write two applications of reluctance motors..
d) Find the slip of a 4-pole three phase induction motor with 50 Hz supply, running at 1460 rpm.
e) Draw the phasor diagram of a salient-pole synchronous motor operating at full-load with lagging power factor
f) Why salient pole construction is rejected for turbo alternator?
g) what is short circuit ratio(SCR)?
h) Delta-delta and Zig-zag transformers are usually used where in a power system.
i) What are Synchronous condensers? Mention its applications.
j) A 400/200V transformer has total resistance of 0.02 p.u on its LV side . what would be its value when it is referred to HV side?

Q2 a) Describe the e.m.f. method and the m.m.f method of determination of voltage regulation of a synchronous generator(cylindrical type). **(5)**

b) A 3-phase, star connected 1000 kVA, 2000 V, 50 Hz synchronous generator gave the open circuit and short circuit test readings as given below. Determine the full load percentage voltage regulation at 0.8 p.f. lag using M.M.F. method. **(5)**

Field current I_f (A)	10	20	25	30	40	50	60
Open circuit line voltage (V)	800	1500	1760	2000	2350	2600	2750
Short circuit armature current (A)		200	250	300			

Q3 a) Explain Blondel's two reaction theory with necessary phasor diagrams. **(5)**

b) A 400V,50Hz, delta-connected alternator has a direct axis reactance of 0.1Ω and a q-axis reactance of 0.07Ω. The armature resistance is negligible. The alternator is supplying 1000A at 0.8 lag p.f.. **(5)**

i. Find the excitation emf neglecting saliency and assuming $X_s=X_{d.}$
ii. Find the excitation emf taking into account the saliency.

Q4 a) Describe the Scott connection and the open delta or V- connection of the three phase transformers. Compare the output of each with a three phase delta connected transformer. **(5)**

b) A three phase transformer 33/6.6 kV, 3MVA has aprimary resistance 8 Ω and secondary resistance of 0.09 Ω per phase.the percentage impedance is 7%. Calculate the secondary load voltage with rated primary voltage and hence the regulation for full load 0.85 p.f lagging condition. **(5)**

Q5 **a)** What do you mean by V curves and inverted V curves? What are causes & effect of it? **(5)**

b) A three phase, salient pole 2300v, 150KW,1000rpm, syn. Motor has Xd = 32Ω ,Xq = 20Ω per phase. Calculate the torque developed if δ = 160, Eb = 2V and neglecting losses. **(5)**

A 100kVA, 11000V,3-ph,Y-connected syn. motor has an effective armature resistance & reactance per phase of 3.5Ω &40Ω respectively. Determine the induced emf& angular retardation of the rotor when fully loaded at

i. Unity p.f
ii. 0.87p.f(lagging)
iii. 0.875p.f(leading)

Q6 **a)** Derive and Explain torque slip characteristic of a three phase IM at running condition. **(5)**

b) A 4 pole,3-phase ,50Hz,27.5 KW inductionmotor is running at full load at 1440 rpm. The star connected rotor has a resistance and a standstill reactance of 0f 0.3 Ω and 2 Ω respectively. The emf between slip ring at stand still is 240 V. Find the induced emf in each rotor phase at full load condition , rotor impedance per phase and rotor current anfp.fassuming the slip rings are short circuited **(5)**

Q7 A 3-ph,Y-connected syn. motor takes 20kW at 400V from the mains. The syn. reactance is 4Ω & the effective resistance is negligible. If the exciting current is so adjusted that the back emf is 550V,calculate the line current and P.f of the motor. **(10)**

Q8 **Write short answer on any TWO :** **(5 x 2)**

a) Explain briefly Reluctance torque of a cylindrical pole synchronous motor.

b) Neatly draw the phasor diagram of a Syn. motor for lagging, unity & leading p f & hence derive the expression for rotor angle(δ).

c) Why parallel operation of alternators are required? Give the necessary conditions for parallel operation of alternators.

d) Explain in details the advantages of Two bright and one dark lamp method over all dark/bright lamp method for syncronisation of an alternator with infinite bus.

Registration No :

Total Number of Pages : 03

Course: B.Tech
Sub_Code: REL5C003

5th Semester Regular/Back Examination: 2022-23
SUBJECT : Electrical Machines-II
BRANCH(S): EE, EEE
Time : 3 Hour
Max Marks : 100
Q.Code : L303

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : **(2 x 10)**

a) Why the pole-shoes of a salient pole synchronous machine are so shaped that their air-gap least in the middle and increases towards the pole edge?

b) What is the effect of 3rd harmonics on a star and delta connected alternator?

c) An alternator was operating at full-load and power factor 0.4 lagging, keeping the load constant the power factor changes to 0.8 lagging. What will the effect on armature reaction?

d) A 3-phase alternator has slots/pole/phase=5. If the coil span=13 slots, determine the 5th harmonics winding factor?

e) What is the condition for maximum power output of a synchronous generator and find the operating power factor during this condition?

f) Why an Induction motor is called as rotating transformer?

g) Compare MMF and synchronous impedance methods of estimating voltage regulation of synchronous alternator.

h) What is synchronizing power of an alternator?

i) What is the function of capacitor in single phase induction motor?

j) What are the applications of synchronous condenser?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 × 8)**

a) Derive the expression for pitch factor and distribution factor?

b) Explain in detail about the concept of Hunting and how to overcome it in a synchronous motor.

c) Explain the torque slip characteristic of 3-phase induction motor.

d) Describe the process of production of rotating magnetic field from a balance three-phase winding.

e) Derive an expression for synchronizing torque when a 3-phase alternator is connected to infinite bus bar.

f) Using double revolving field theory, explain why a single-phase induction motor is not self-starting.

g) Explain the process of determining the parameters of a single-phase induction motor by oc and sc test.

h) Discuss the phenomena of Cogging and Crawling in an induction motor.

i) Describe the starting of an induction motor by start-delta and autotransformer starting method.

j) Explain how potier triangle can be drawn, and the process of calculating voltage regulation by zpf method from potier triangle?

k) Proof that $P_{tm} - P_{em} = I_a^2 R_a$.

l) Explain the effect of increasing excitation of one of the alternators when two alternators are connected in parallel.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 a) A 6 pole, 50 Hz, 3-Phase, induction motor running on full load develops a useful torque of 160 N m. When the rotor emf makes 120 complete cycle per minute. Calculate the shaft power input. If the mechanical torque lost in friction and that for core loss is 10 N m. Considering a total stator loss of 800 W, compute: **(10+6)**

i) The copper loss in the rotor windings
ii) The input the motor.
iii) The efficiency.

b) A 3-phase induction motor having a 6pole, star connected stator runs from 240V, 50HZ supply .The rotor resistance and standstill reactance are 0.12 Ω and 0.85 Ω per phase .The ratio of stator to rotor turns is 1.8 and full load slip is 4%. Calculate the developed torque at full load and maximum torque. Also calculate the slip at which maximum torque occurs.

Q4 a) A 3-phase, 4-pole, 50 Hz, induction motor has a star connected wound rotor. **(10+6)**
The rotor emf is 50V between the slip rings at standstill. The rotor resistance
and standstill reactance are 0.4 Ω and 2.0 Ω respectively. Calculate

i) rotor current per phase at starting when slip rings are short circuited
ii) Rotor current per phase at starting if 50 Ω per phase resistance is connected between slip rings.
iii) Rotor EMF when the motor us running at full load at 1440 rpm
iv) Rotor current at full load and
v) Rotor power factor at full load

b) A 15 Hp, three-phase, 6 pole, 50 Hz, 400 V, delta connected IM runs at 960 rpm on full load. If it takes 86.4 A on direct starting, find the ratio of starting torque to full-load torque with a star- delta starter. Full load efficiency and power factor are 88% and 0.85 respectively.

Q5 **a)** A 3-phase, 1.5 MVA, 11 KV star connected alternator has synchronous impedance of 1.2+j25 Ω per phase. Find the full load voltage regulation and load angle of operation at pf 0.8 leading. Also, calculate the power factor at which the voltage regulation will be zero? **(8+8)**

b) Two alternators A and B operate in parallel and supply a load of 10MW at 0.8pf
lagging. (i) By adjusting steam supply of A, its power output is adjusted to 6,000KW and by changing its excitation, its p.f is adjusted to 0.92 lag. Find the
Power Factor of alternator B. (ii) If steam supply of both machines is left unchanged, but excitation of B is reduced so that it's P.F becomes 0.92 lead, find new p.f of A.

Q6 **a)** A 3.3 kV, star connected synchronous motor has per phase impedance of 0.4+j5 Ω. The motor draws 1000 kW with an excitation emf of 4 kV. Compute the line current and power factor? **(8+8)**

b) A 3-phase, 4 pole, 1500 rpm alternator has diameter of 30 cm, length=25 cm, and double layer winding. The winding is accommodated in 4 slots/pole/phase with 5 turns/coil. The coils are short pitched by 1 slot. The fundamental flux density is 0.87 T, whereas third is 0.23T. Determine line and Phase voltages when connected in star?

Registration No: -

Total Number of Pages:02 **B. Tech**

REL5C003

5th Semester Regular / Back Examination: 2021-22

ELECTRICAL MACHINES-II

Branch: EEE, ELECTRICAL

Max Marks: 100

Time: 3 Hours

Q Code: OF308

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) **(02×10)**

a) State two advantages of short pitched coils in the armature winding in an AC machine. **(2)**

b) Explain why load angle 'delta' is positive for synchronous generator & negative for in case of synchronous motor? **(2)**

c) Determine the distribution factor of a 3 phase balanced distributed winding having 9 slots/pole & a coil span of 8 slots. **(2)**

d) An alternator has the same voltage, frequency and phase sequence as that of the infinite bus. Is it sufficient to put ON the synchronization switch now? Give reasons for your answer? **(2)**

e) What do you understand by synchronization of alternators? **(2)**

f) Why per phase X_d of a salient pole three-phase alternator is greater than its X_q? **(2)**

g) For the same kVA load, efficiency of an alternator at leading p.f. is more than at lagging p.f. Explain. **(2)**

h) Explain the objective of using an auxiliary winding in a single-phase induction motor? How it is different from main winding? **(2)**

i) What is the speed regulation of a synchronous motor ?Justify. **(2)**

j) A 50 Hz and 60 Hz alternator are mechanically coupled together. At what maximum speed it should rotate so as to induce voltage at their rated frequency? **(2)**

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(06×08)**

a) A 3-phase, 6 pole, 1000 rpm alternator has diameter of 28 cm, length=23 cm, and double layer winding. The winding is accommodated in 4 slots/pole/phase with 4 turns/coil. The coils are short pitched by 1 slot. The fundamental flux density is 0.87T, whereas third and fifth are B_3=0.23T and B_5=0.14T. Calculate line and Phase voltages when connected in star. **(6)**

b) A 3-phase star connected synchronous motor has synchronous impedance of 1+j 10 Ω per phase. It is operating at a constant voltage of 6.6kV and its field current is adjusted to give an excitation voltage of 6.4kV. Find the power output, armature current and pf under the condition of maximum power output. **(6)**

c) What do you understand by parallel operation of Alternators? Explain different conditions required for parallel operation of three phase synchronous generator to infinite bus ? **(6)**

d) Draw and explain the equivalent circuit and phasor diagram of three phase induction motor. **(6)**

e) Differentiate between three phase and single phase induction motor. Explain double revolving field theory for single phase induction motor. **(6)**

f) A 10 kW, 400 V, 4 pole delta connected squirrel cage induction motor gave the following test results: **(6)**

No load test: 400V, 8A, 250W

Blocked rotor test: 90V, 35A, 1350W

The DC resistance of the stator winding per phase measured immediately after the blocked rotor test is 0.6ohm. Calculate the rotational losses and the equivalent circuit parameters.

g) Analyze how armature reaction in synchronous generator is going to affect its generated EMF. **(6)**

h) Explain the synchronous impedance method of voltage regulation calculation of Alternator. Discuss its advantages and disadvantages. **(6)**

i) A 3-phase, 20kVA, 400 V, star connected alternator is delivering rated load at 400 V and at pf 0.8 lagging. Its synchronous impedance is 0.5+j4 Ω per phase. Find the load angle of operation and voltage regulation. Also calculate the p.f. for which the voltage regulation becomes zero. **(6)**

j) Two similar alternators operate in parallel have the following data: **(6)**

Alternator 1: 700kW, frequency drops from 50Hz at no load to 48.5 Hz at full load.

Alternator 2: 700kW, frequency drops from 50.5Hz at no load to 48 Hz at full load.

Determine (i) how a total load of 1200kW is shared between two and the operating frequency, (ii) maximum load that can be delivered by both without overloading either of them.

k) A 440V, 50Hz, 6-pole, Delta connected 3-phase induction motor consumes 45 kW with a line current of 75A and runs at a slip of 3%. If stator iron loss is 1200W and friction loss is 900W and resistance between two stator terminals is 0.12ohm,calculate (i) power supplied to the rotor (ii) rotor Cu loss (iii) power supplied to load (iv) efficiency and (v)shaft torque developed. **(6)**

l) Explain different methods of starting of three phase induction motor. **(6)**

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four) **(02×16)**

Q3 Discuss various methods of starting used for Synchronous motor. Explain with the help of phasor diagram , the effect of varying excitation on armature current and power factor in a synchronous motor ? **(16)**

Q4 Discuss two reaction theory of salient pole synchronous machine and hence explain its phasor diagram. Compare it with cylindrical rotor synchronous machine. **(16)**

Q5 Describe various methods for controlling the speed of a 3 phase induction motor. Also explain slip power recovery method for speed control ? **(16)**

Q6 Draw the circuit model of a single phase induction motor, and explain how the parameters can be calculated from no load and blocked rotor test. **(16)**

Registration No :

Total Number of Pages : 02 **B.Tech**

PCEL4302

5th Semester Back Examination 2019-20

ELECTRICAL MACHINES - II

BRANCH : EEE, ELECTRICAL

Time : 3 Hours

Max Marks : 70

Q.CODE : HB159

Answer Question No.1 which is compulsory and any FIVE from the rest.

The figures in the right hand margin indicate marks.

Q1 **Answer the following questions :** **(2 x 10)**

- **a)** Explain with example, how coil pitch affects induced emf in a three phase synchronous generator?
- **b)** State the conditions for performing open circuit and short circuit tests on a three phase synchronous generator.
- **c)** Draw the power angle characteristics for a three phase synchronous generator and explain the importance of the curve.
- **d)** Write two important requirements for operating an isolated three phase synchronous generator in parallel with an infinite busbar.
- **e)** How is it possible to change the active power generation of athree phase synchronous generator? Give example of any device which may perform this requirement.
- **f)** State the mechanism of starting a three phase synchronous motor.
- **g)** Write two important differences that you may observe in the construction of salient pole type three phase synchronous generator and cylindrical rotor type three phase synchronous generator.
- **h)** Why the transmission and distribution networks require installation of transformers in the power system?
- **i)** Draw the single line connection diagram for a delta-delta and a delta-star three phase transformer illustrating smooth transmission and distribution of electric power in a practical network.
- **j)** Determine the suitable number of poles required for a three phase synchronous generator to produce output at 50Hz, if the turbine speed is 3000 rpm.

Q2 **a)** Draw the equivalent circuit of a three phase synchronous generator, and analyze its performance for leading power factor operation with the help of a suitable phasor diagram. **(5)**

b) A three phase synchronous generator supplies rated power at 0.8 p.f. lagging. If the resistance is 0.01 pu, reactance is 1 p.u., and excitation voltage is 2 p.u., determine the possible terminal voltage of the machine. **(5)**

Q3 **a)** During the open circuit test on a three phase star connected synchronous generator with ratings of 50 Hz, 2000 V, a field current of 30 A produces an OC terminal voltage of 1000 V across the lines. Also, a full load current of 150 A is recorded when a short circuit test is performed on the same machine having an effective resistance of 0.4 ohm per phase. Determine; **(5)**

i) Synchronous impedance of the machine

ii) Regulation of the machine at 0.9 pf lagging

b) Justify in the context of armature reaction that the effect of armature mmf on main field mmf is demagnetizing in nature for the case of an alternator and magnetizing in nature for that of a synchronous motor. **(5)**

Q4 a) Derive the torque expression for both cylindrical and salient pole type three phase synchronous motors and give a clear comparison for the similarities and differences, of the two case, on the basis of the torque expression. **(5)**

b) A 400 V, 3 phase, star connected synchronous motor is rated for 100 hp. If the resistance and leakage reactance of the machine are 0.03 ohm and 0.5 ohm respectively, Calculate the efficiency and open circuit emf per phase. Assume that the machine is operating at 0.8 leading p.f. while the mechanical power developed by it is 80 KW. **(5)**

Q5 a) Draw the connection diagram and phasor diagrams for the following types of three phase transformer connections. **(5)**

i) Yy0
ii) Dd6
iii) Yd11

b) Justify, whether a three phase delta-star transformer (T1) with delta connection at the primary side can be operated in parallel with another three phase star-delta transformer (T2) whose primary is star connected? If yes, determine the ratio (*x:y*), assuming that the primary to secondary turns ratio of T1 is '*x*' and the primary to secondary turns ratio of T2 is '*y*'. **(5)**

Q6 Describe the process of finding the expression for voltage regulation of a three phase synchronous generator by using the following methods, and hence compare them. **(10)**

i) synchronous impedance method
ii) zero power factor method

Q7 a) Discuss the two reaction concept as regards to the development of rotating magnetic field. **(5)**

b) Justify the need for finding the direct axis and quadrature axis reactance separately for the case of salient pole type alternators. Also describe how to evaluate the direct axis and quadrature axis reactance for the same machine by performing slip test. **(5)**

Q8 **Write short Notes on any TWO :** **(5 x 2)**

a) Importance of Power angle Equation and Power angle Characteristics
b) Load sharing mechanism during operation of synchronous generators in parallel
c) Speed control of single phase induction motor